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Sign Language Translation

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Abstract

Sign Language Translation represents a crucial technology aimed at bridging the communication gap between deaf and hearing communities by automatically translating sign language into written language. With sign languages being the primary means of communication for millions of deaf and hard-of-hearing individuals worldwide, our work focuses on developing an automated system for translating sign language into natural language. To the best of our knowledge, this project represents one of the early efforts in exploring this translation direction within our target setting. By accurately interpreting visual gestures and hand movements, our system enhances accessibility, promotes inclusivity, and facilitates seamless communication across diverse social and cultural contexts. This work contributes to breaking communication barriers and advancing assistive technologies, enabling broader participation of the deaf community in education, media, and daily interactions.

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List of Acronyms

Adp Msk	Adaptive Masking
ASL	American Sign Language
AVRET	Adaptive Video Representation Enhanced Transformer
BLEU	Bilingual Evaluation Understudy
CSL	Chinese Sign Language
CTC	Connectionist Temporal Classification
DGS	Deutsche Gebärdensprache (German Sign Language)
EASLT	Emotion-Aware Sign Language Translation
FDA	Frame Data Augmentation
ISLR	Isolated Sign Language Recognition
LLM	Large Language Model
RGB	Red, Green, Blue (Color Model)
RRC	Reproducible Random Context
SLR	Sign Language Recognition
SLT	Sign Language Translation
SpaMo	Spatial and Motion-based Sign Language Translation
SSCL	Spatial-Spatiotemporal Contrastive Loss
TCN	Temporal Convolutional Network
VT Sep	Visual Text Separator

Chapter 1

Introduction

1.1 Background

Sign languages are natural languages that use visual-spatial modalities such as hand shape, movement, facial expressions, and body posture to convey meaning. Unlike spoken languages, sign languages are inherently multimodal and highly temporal, making automatic translation a challenging problem. Sign Language Translation (SLT) aims to convert sign language videos directly into spoken language text, enabling effective communication between Deaf and hearing communities.

Early SLT systems relied heavily on gloss annotations, which are symbolic word-level representations of signs. However, producing gloss annotations is expensive, subjective, and does not fully capture the linguistic richness of sign languages. This has motivated recent research toward gloss-free SLT, where models learn translations directly from visual input without requiring intermediate gloss supervision.

With the emergence of powerful deep learning models, particularly Transformer-based architectures and Large Language Models (LLMs), gloss-free SLT has shown promising results. Nevertheless, significant challenges remain in learning robust visual representations, modeling long temporal dependencies, and effectively integrating contextual information.

1.2 Problem Statement and Motivation

Despite recent progress, current gloss-free SLT systems still struggle with several fundamental issues: Representation Density: Visually similar signs with different meanings are often mapped to nearby feature representations, leading to ambiguity in translation. Rigid Temporal Modeling: Fixed temporal receptive fields fail to align with the variable duration and boundaries of signs. Sensitivity to Visual Noise: Changes in background, lighting, or signer appearance can significantly degrade performance. Underutilization

of Context: Many models rely solely on visual features and ignore valuable linguistic or environmental context that could disambiguate translations.

These limitations directly impact translation quality, especially in real-world scenarios. The motivation of this project is to address these challenges by enhancing representation learning, improving robustness, and enabling better use of contextual cues, while maintaining a fully gloss-free pipeline.

1.3 Report Organization

This report follows a structured outline:

Chapter 1 introduced the project, discussed its motivation, outlined the problem we are addressing, and provided a roadmap for the report.

Chapter 2 reviews previous models and solutions published in top conferences and journals, such as CVPR.

Chapter 3 presents the methodology and conducted work from the first academic term.

Chapter 4 details the refined methodology and structural enhancements introduced during the second academic term.

Chapter 5 provides a comprehensive conclusion of the project and outlines potential directions for future research.

Chapter 2

Literature Review

2.1 Introduction

In this chapter, we present a comprehensive overview of prominent research contributions in the field of Gloss-Free Sign Language Translation (SLT).

The remainder of this chapter is organized as follows. Section 2.2 reviews previously proposed models and methodologies for sign language translation, with a focus on advancements in visual feature extraction, the introduction of loss functions for improved representation learning, and the integration of multi-modal contextual information. Section 2.3 introduces the datasets utilized in this work, highlighting their relevance, characteristics, and limitations in the context of sign language translation.

2.2 Previously Proposed Models

2.2.1 Multilingual Gloss-free Sign Language Translation: Towards Building a Sign Language Foundation Model [1]

Recent studies have demonstrated that the choice of visual encoder plays a critical role in performance. This work highlights the importance of domain-specific pre-training on large-scale Sign Language Recognition (SLR) datasets. To address the limitations of general-purpose backbones, this approach leverages architectures such as the Video Swin Transformer, which models long-range spatio-temporal dependencies. To solve the challenge of learning shared representations across languages, the model employs a hierarchical encoder trained with dual Connectionist Temporal Classification (CTC) objectives. One CTC loss is used for sign language identification, while a second text-oriented CTC loss is applied to the generated spoken text.

2.2.2 Lost in Translation, Found in Context: Sign Language Translation with Contextual Cues [2]

To address ambiguities in visual-only SLT (e.g., homonyms or pointing gestures), this framework integrates external contextual cues. The authors augment a pre-trained LLM with two specific textual inputs: pseudo-glosses generated by a pre-trained Isolated Sign Language Recognition (ISLR) classifier, and background descriptions from an image captioning model. To ensure robustness, the authors apply specific Input Augmentation Strategies. "Word Dropping" randomly omits words within cues to prevent overfitting, while "Cue Dropping" discards entire inputs to force the model to function when modalities are missing.

2.2.3 Sign2GPT: Leveraging Large Language Models for Gloss-Free SLT [3]

This paper bridges the modality gap using a Pseudo-Gloss Context mechanism during pre-training. Unlike ISLR-based methods, this approach generates pseudo-glosses from spoken sentences by filtering for semantically relevant parts of speech, such as nouns and verbs. These pseudo-glosses act as prototypes, allowing the model to learn visual-linguistic mappings implicitly without expensive manual gloss annotations.

2.2.4 Improving End-to-end SLT with Adaptive Video Representation [4]

Temporal alignment is a long-standing challenge, as fixed-size sliding windows often fail to align with natural sign boundaries. The AVRET model addresses this through a dynamic clip partitioning strategy, which enriches video segments with contextually important frames from neighboring clips using learned scoring mechanisms. Complementing this is the Adaptive Fusion mechanism. Instead of simple concatenation, this module calculates adaptive weights to fuse spatio-temporal features from the transformer encoder with continuous temporal features generated by an Adaptive Masking module.

2.2.5 Gloss Attention for Gloss-free Sign Language Translation [5]

Temporal modeling often relies on TCNs, which impose a fixed receptive field. To address this limitation, this paper proposes a gloss-informed attention mechanism that dynamically adjusts its focus based on learned sign-level representations, effectively approximating the role of explicit gloss boundaries. In addition, this work introduces

a knowledge transfer strategy to align visual representations with the outputs of a powerful pre-trained language model, such as Sentence-BERT, enforcing a globally coherent semantic structure within the visual encoder.

2.2.6 Improving Gloss-free Sign Language Translation by Reducing Representation Density [6]

A critical limitation in gloss-free SLT is the "representation density problem," where visually similar but semantically different signs are encoded into densely packed regions in the feature space. To address this, the authors propose Sign Contrastive Learning (SignCL). SignCL is a self-supervised objective integrated into training that encourages representations of sign segments with similar semantics to be closer in the embedding space, while enforcing separation between semantically different signs.

2.2.7 Uni-Sign: Unified Framework for Sign Language Understanding [7]

While RGB-based visual features capture appearance, they are sensitive to environmental variations such as illumination and cluttered backgrounds. To mitigate these limitations, Uni-Sign proposes a unified framework that explicitly incorporates human pose representations derived from keypoint detection models. By encoding joint positions and motion trajectories, pose-based features provide a compact and noise-robust description of hand, arm, and upper-body movements.

2.2.8 An Efficient Gloss-Free Sign Language Translation Using Spatial Configurations and Motion Dynamics with LLMs (SpaMo) [8]

Conventional gloss-free methods often rely on computationally expensive, domain-specific fine-tuning of visual encoders to achieve optimal results. To overcome this, the SpaMo framework proposes a novel approach that explicitly models two fundamental linguistic characteristics of sign language: *spatial configurations* and *motion dynamics*.

Instead of fine-tuning, this model utilizes off-the-shelf, frozen visual encoders to extract these features separately. A pre-trained image encoder captures spatial details (hand shapes, posture), while a video encoder captures temporal dynamics. These features are then fused and aligned with a Large Language Model (LLM) using a lightweight projector. To bridge the modality gap, SpaMo introduces a *Visual-Text*

Alignment (VT-Align) warm-up stage, achieving state-of-the-art performance without the need for extensive visual backbone training.

2.2.9 EASLT: Emotion-Aware Sign Language Translation [9]

Current gloss-free SLT methods often focus entirely on hand gestures and ignore facial expressions, which creates ambiguity when different signs share the exact same hand movements. To address this issue, the EASLT framework treats facial expressions as a primary feature rather than just an afterthought by processing the video in three separate ways at the same time: it tracks what the hands look like (spatial features), how they move (motion features), and how the signer’s facial expressions change (emotional features). A special component called the Emotion-Aware Fusion (EAF) module then brings these three pieces together, using the facial expressions as clues to better interpret the hand movements. Because the system actively uses the signer’s face for context, it stops getting confused by hand signs that look identical, making the final translation much more accurate.

2.2.10 SignMouth: Leveraging Mouthing Cues for Sign Language Translation by Multimodal Contrastive Fusion [10]

SignMouth focuses on integrating non-manual mouthing cues with manual gesture features to disambiguate visually similar signs. The paper proposes a multimodal contrastive fusion strategy that aligns lip/mouth movement features with spatial gesture representations and textual targets via hierarchical contrastive objectives. These multi-level alignment losses enforce consistency across sign-lip and visual-text modalities, improving robustness to visual noise and signer variability. Extensive experiments on PHOENIX14T and How2Sign show consistent gains over prior gloss-free baselines, demonstrating the value of explicit mouthing cues in SLT.

2.3 Datasets

In this section, we delve into the discussion of the primary dataset utilized in our sign language translation task. The scarcity of high-quality, annotated datasets for sign language has been a long-standing challenge in the field.

2.3.1 PHOENIX14T [11]

The RWTH-PHOENIX-Weather 2014 T (PHOENIX14T) dataset is the most widely used benchmark for Sign Language Translation (SLT). It is derived from the German

public television station PHOENIX, specifically featuring weather forecasts interpreted into German Sign Language (DGS). Unlike other datasets that may focus on isolated signs, PHOENIX14T is designed for continuous sign language translation. It contains parallel video, gloss annotations, and German spoken text translations. The dataset consists of approximately 8,257 videos performed by 9 different signers.

2.3.2 CSL-Daily [12]

CSL-Daily is a large-scale dataset for continuous Sign Language Translation collected in a real-world daily communication setting. It focuses on Chinese Sign Language (CSL) and contains sign language videos paired with spoken language sentences, without relying on constrained domains such as weather forecasting.

The dataset consists of over 20,000 video-sentence pairs performed by multiple signers, covering a wide range of daily-life topics and linguistic expressions. Compared to PHOENIX14T, CSL-Daily exhibits higher vocabulary diversity, longer sentences, and more complex syntactic structures, making it particularly suitable for evaluating the scalability and generalization ability of SLT models.

Due to its size and linguistic variability, CSL-Daily has been widely used to study data-efficient learning, cross-domain robustness, and the impact of additional supervision techniques such as back-translation and contrastive learning.

2.3.3 How2Sign [13]

How2Sign is a large-scale multimodal dataset designed for continuous American Sign Language (ASL) understanding and translation. The dataset contains professionally recorded sign language videos aligned with spoken English sentences, gloss annotations, audio, and multiple camera views.

Unlike broadcast-based datasets, How2Sign is recorded in controlled environments with consistent lighting and camera setups, while still preserving natural signing styles. It includes a diverse set of speakers and sentence structures, making it well-suited for studying long-range temporal modeling and multimodal alignment.

The richness of annotations and modalities in How2Sign enables advanced research on cross-modal representation learning, large-scale training of SLT systems, and evaluation of model robustness across different sign languages and domains.

Chapter 3

First Term: Used Approach and Conducted Work

3.1 Introduction

This chapter presents the conducted work and the adopted methodology for building our gloss-free Sign Language Translation (SLT) system. The proposed system is based on the Spatial and Motion-based Sign Language Translation (SpaMo) framework [8], which explicitly models spatial configurations and motion dynamics using off-the-shelf visual encoders without domain-specific fine-tuning. In addition, several enhancements inspired by recent literature are integrated to improve robustness, representation quality, and translation performance.

3.2 Model Architecture

3.2.1 Overview

The proposed system follows a fully end-to-end, gloss-free translation pipeline. It decouples spatial and temporal information from sign videos and aligns the extracted visual representations with a large language model (LLM) for sequence generation.

SpaMo focuses on two fundamental linguistic characteristics of sign language: *spatial configurations* and *motion dynamics*. Spatial configurations describe hand shapes, facial expressions, and body posture within the signing space, while motion dynamics capture the trajectory, rhythm, and temporal evolution of movements. Modeling these aspects separately enables more discriminative and robust representations.

Figure 3.1 illustrates the overall architecture of the SpaMo-based system used in this work.

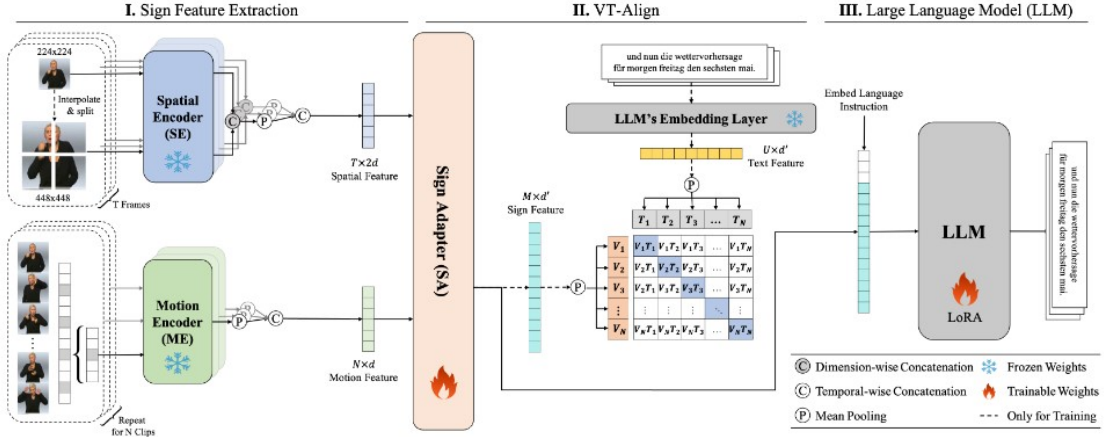


Figure 3.1: Overview of the SpaMo-based Sign Language Translation framework.

3.2.2 Pipeline

Given an input sign video, the processing pipeline consists of the following stages:

1. **Spatial Feature Extraction:** Each frame is independently encoded using a frozen pre-trained image encoder.
2. **Motion Feature Extraction:** Overlapping temporal clips are processed using a frozen pre-trained video encoder to capture motion dynamics.
3. **Feature Integration:** Spatial and motion features are unified using a Sign Adapter module that aligns them with the language model embedding space.
4. **Visual-Text Alignment:** A contrastive warm-up phase aligns visual representations with textual embeddings before full translation training.
5. **Sequence Generation:** A T5-based language model generates the target spoken language sentence.

3.2.3 Spatial Encoder

The Spatial Encoder extracts frame-level spatial representations using a pre-trained image encoder such as Vision Transformer (ViT), which remains frozen throughout training. To enhance spatial richness, Scaling on Scales (S2) is applied, enabling multi-scale feature extraction without modifying encoder parameters. This allows both fine-grained and global spatial details to be captured.

3.2.4 Motion Encoder

The Motion Encoder captures temporal dynamics using a frozen pre-trained video encoder such as VideoMAE. Since explicit gloss boundaries are unavailable in gloss-

free settings, a sliding window strategy is employed to generate implicit gloss-level motion representations. This approach enables modeling of rhythm, repetition, and temporal variation inherent in sign language.

3.2.5 Sign Adapter and Visual-Text Alignment

Spatial and motion features differ in dimensionality and temporal resolution. To integrate them effectively, a Sign Adapter module is employed, consisting of linear projections, a temporal convolutional network (TCN), and a multi-layer perceptron (MLP). The output is a unified sign representation aligned with the embedding space of the language model.

To further reduce the modality gap, Visual-Text Alignment (VT-Align) is applied as a warm-up training strategy based on contrastive learning. VT-Align initializes the Sign Adapter with semantically meaningful weights before full SLT supervision, leading to more stable and effective training.

3.2.6 Experimenting with Large Language Models

In the original SpaMo framework, text generation is performed using FLAN-T5-XL, which provides strong sequence modeling and generative capabilities [8]. However, training and fine-tuning such large-scale models require substantial computational resources.

Due to resource constraints associated with running experiments on Kaggle, where training is limited to two NVIDIA T4 GPUs with restricted memory and runtime, the SpaMo architecture is preserved in its entirety with the exception of the language model component. Instead of FLAN-T5-XL, we experiment with a series of smaller instruction-tuned T5-based models, including FLAN-T5-Small, FLAN-T5-Base, and FLAN-T5-Large.

To enable efficient adaptation while maintaining the pre-trained linguistic knowledge of these models, we employ Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning. Under this setup, only a small subset of parameters is updated during training, significantly reducing memory usage and computational cost.

We noticed a non-monotonic pattern with Flan-T5, as shown in Figure 3.2, where the Small and Base models achieved similar performance, the Large model performed significantly worse, and the XL model achieved the best performance among all variants. This behavior corresponds to the deep double descent phenomenon and was triggered by the inclusion of the context feature (the output sentence translation). Since Flan-T5 already is a translation-tuned LLM, the context feature acted as a dominant signal, causing the model to ignore visual features.

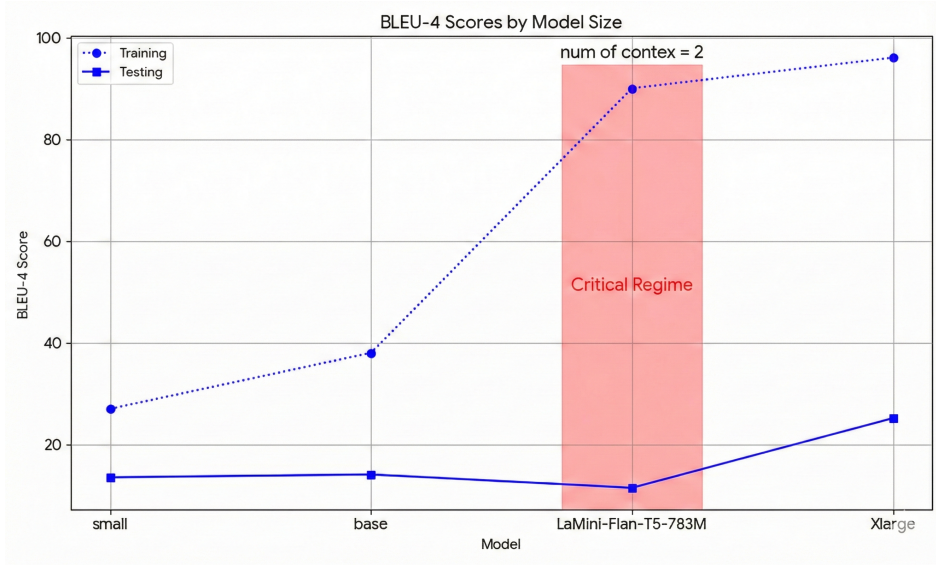


Figure 3.2: Impact of model size on performance showing deep double descent.

After removing context feature, the issue was resolved, and performance improved monotonically with increasing model size, as detailed in Table 3.1.

Table 3.1: Baseline performance comparison.

Metric	Small	Base	Large
BLEU-1	38.30	40.26	40.89
BLEU-2	24.88	26.55	27.09
BLEU-3	17.76	19.28	19.77
BLEU-4	13.53	15.02	15.46
ROUGE-L (F1)	0.33	0.36	0.36
ROUGE-L (P)	0.34	0.36	0.37
ROUGE-L (R)	0.33	0.36	0.36
Contrastive Loss	0.24	0.33	0.37

3.3 Conducted Work

3.3.1 Pose Features

We incorporate pose information extracted via MediaPipe to capture fine-grained manual and non-manual details often missed by standard visual embeddings. This data helps the model to distinguish visually similar signs.

As shown in Table 3.2, the addition of geometric guidance consistently improves translation quality across the **Small**, **Base**, and **Large** flan-t5 models.

Table 3.2: Performance with pose feature integration.

Metric	Small		Base		Large	
	<i>Base</i>	<i>+Pose</i>	<i>Base</i>	<i>+Pose</i>	<i>Base</i>	<i>+Pose</i>
BLEU-1	38.30	39.19	40.26	40.12	40.89	40.31
BLEU-2	24.88	26.33	26.55	27.00	27.09	27.28
BLEU-3	17.76	19.23	19.28	20.04	19.77	20.27
BLEU-4	13.53	15.12	15.02	15.90	15.46	16.06
ROUGE-L (F1)	0.33	0.34	0.36	0.34	0.36	0.35
ROUGE-L (P)	0.34	0.35	0.36	0.35	0.37	0.36
ROUGE-L (R)	0.33	0.34	0.36	0.35	0.36	0.35
Contrastive Loss	0.24	0.47	0.33	0.37	0.37	0.43

3.3.2 Adaptive Fusion

Adaptive Fusion to dynamically weigh spatial, spatiotemporal, and pose inputs.

Table 3.3: Performance with adaptive fusion feature integration.

Metric	Small		Base		Large	
	<i>Base</i>	<i>+Adaptive</i>	<i>Base</i>	<i>+Adaptive</i>	<i>Base</i>	<i>+Adaptive</i>
BLEU-1	38.30	37.87	40.26	39.30	40.89	37.95
BLEU-2	24.88	24.37	26.55	25.76	27.09	24.91
BLEU-3	17.76	17.57	19.28	18.69	19.77	18.21
BLEU-4	13.53	13.68	15.02	14.61	15.46	14.44
ROUGE-L (F1)	0.33	0.32	0.36	0.34	0.36	0.33
ROUGE-L (P)	0.34	0.33	0.36	0.34	0.37	0.34
ROUGE-L (R)	0.33	0.32	0.36	0.34	0.36	0.33
Contrastive Loss	0.24	0.52	0.33	0.45	0.37	0.46

The results in Table 3.3 show no improvement. Unlike the AVRET framework, which fuses two compatible visual representations (Transformer and BiLSTM embeddings), our method fused different modalities (visual embeddings and geometric coordinates). This discrepancy prevented the model from effectively combining the features, so we will decline this feature.

3.3.3 Data Augmentation

Inspired by the robustness-oriented augmentation strategies proposed in [2], temporal and feature-level augmentations are applied to visual embeddings during training. Frame dropout reduces sensitivity to exact temporal alignment, channel dropout prevents over-reliance on specific feature dimensions, and span masking encourages global temporal reasoning. Although the implementation differs, the applied augmentations are conceptually motivated by the same regularization principles.

As shown in Table 3.4, The improvements observed with the Small and Base models,

Table 3.4: Performance impact of Data Augmentation.

Metric	Small		Base		Large	
	<i>Base</i>	<i>+DA</i>	<i>Base</i>	<i>+DA</i>	<i>Base</i>	<i>+DA</i>
BLEU-1	38.30	38.31	40.26	40.29	40.89	39.52
BLEU-2	24.88	25.23	26.55	27.03	27.09	26.19
BLEU-3	17.77	18.55	19.28	19.87	19.77	19.18
BLEU-4	13.53	14.73	15.02	15.63	15.46	15.18
ROUGE-L (F1)	0.33	0.33	0.36	0.35	0.36	0.34
ROUGE-L (P)	0.35	0.34	0.36	0.36	0.37	0.35
ROUGE-L (R)	0.33	0.33	0.36	0.35	0.36	0.34
Contrastive Loss	0.25	0.38	0.33	0.33	0.37	0.36

but not with the Large variant, can be attributed to the fact that the small, and base models exhibited a significantly higher tendency to overfit under the used parameters compared to the Large model. Consequently, the fixed data augmentation probabilities (e.g., frame, span, and channel dropout rates) acted as an effective regularizer that mitigated this specific overfitting for the Small and Base variants. In contrast, for the Large model, which was less prone to this overfitting regime, these same augmentations likely introduced excessive perturbations rather than beneficial regularization. As a result, the augmentation strategy likely requires scale-aware re-tuning to better match the representational capacity of larger FLAN-T5 variants.

3.3.4 Sign Contrastive Learning

To mitigate representation density in gloss-free SLT, Sign Contrastive Learning (SignCL) is incorporated following [6]. This auxiliary objective encourages semantically similar signs to form compact clusters in the embedding space while enforcing separation between dissimilar signs, improving semantic discrimination. We explore different configurations of the SignCL objective by varying the loss weight α , temperature, step size, and temporal window size.

As shown in Table 3.5, the integration of SignCL leads to a consistent decrease in BLEU-1 across the Small, Base, and Large models. In contrast, higher-order n-gram metrics (BLEU-3 and BLEU-4) improve for the Small and Base models, indicating better capture of longer-range dependencies and sentence-level structure.

This behavior reflects a trade-off between token-level accuracy and sequence-level coherence. While BLEU-1 emphasizes individual word matching, BLEU-4 is more sensitive to fluent and structurally consistent translations. Larger values of the loss weight α combined with low temperatures amplify the influence of the contrastive objective, encouraging semantic alignment at the cost of exact token prediction.

By adopting more moderate hyperparameter settings, specifically reducing α and increasing the temperature, we obtain a better balance between semantic discrimina-

Table 3.5: Performance with SignCL integration.

Metric	Small		Base		Large	
	<i>Base</i>	<i>+SignCL</i>	<i>Base</i>	<i>+SignCL</i>	<i>Base</i>	<i>+SignCL</i>
BLEU-1	38.30	37.78	40.26	40.03	40.89	39.54
BLEU-2	24.88	24.64	26.55	27.00	27.09	26.3
BLEU-3	17.76	17.87	19.28	19.92	19.77	19.29
BLEU-4	13.53	13.96	15.02	15.72	15.46	15.09
ROUGE-L (F1)	0.33	0.322	0.36	0.35	0.36	0.34
ROUGE-L (P)	0.34	0.33	0.36	0.36	0.37	0.35
ROUGE-L (R)	0.33	0.32	0.36	0.35	0.36	0.35
Contrastive Loss	0.24	0.45	0.33	0.39	0.37	0.37
SignCL Loss	–	0.44	–	0.49	–	0.47

tion and sequential consistency. The Large model does not exhibit comparable gains, which we attribute to the reuse of SignCL hyperparameters tuned on the Small model without capacity-specific optimization. Due to time and computational constraints, these settings were directly transferred to the Base and Large variants; further tuning is left for future work.

3.4 Datasets Used

All experiments in this work are conducted on the RWTH-PHOENIX-Weather 2014 T (PHOENIX14T) dataset [11], which consists of continuous German Sign Language videos paired with spoken language translations. PHOENIX14T is the most widely used benchmark for evaluating gloss-free SLT systems.

3.5 Experimental Results

3.5.1 Ablation Study

We conduct an ablation study to analyze the individual and combined effects of the proposed enhancements on the SpaMo-based system. Results are reported on the PHOENIX14T validation set for FLAN-T5-Small, FLAN-T5-Base, and FLAN-T5-Large language models.

Table 3.6: Ablation study for FLAN-T5-Small on PHOENIX14T (Validation).

Configuration	B1	B2	B3	B4	RL-F1	RL-P	RL-R	Contra	SignCL	Warmup
Original	38.30	24.88	17.76	13.53	0.33	0.34	0.33	0.24	–	–
Pose	39.19	26.33	19.23	15.12	0.34	0.35	0.34	0.47	–	–
SignCL	37.78	24.64	17.87	13.96	0.32	0.33	0.32	0.45	0.44	–
DA	38.31	25.23	18.55	14.73	0.33	0.34	0.33	0.38	–	–
Pose + DA	38.49	25.72	19.08	15.22	0.34	0.35	0.34	0.36	–	–
Pose + SignCL	37.78	24.64	17.87	13.97	0.32	0.33	0.32	0.45	0.44	0.49
Pose + SignCL + DA	39.34	26.04	18.92	14.83	0.34	0.35	0.34	0.37	0.42	0.42

Table 3.7: Ablation study for FLAN-T5-Base on PHOENIX14T (Validation).

Configuration	B1	B2	B3	B4	RL-F1	RL-P	RL-R	Contra	SignCL	Warmup
Original	40.26	26.55	19.28	15.02	0.36	0.36	0.36	0.33	–	–
Pose	40.12	27.00	20.04	15.90	0.34	0.35	0.35	0.37	–	–
SignCL	40.03	27.00	19.92	15.72	0.35	0.36	0.35	0.39	0.49	–
DA	40.29	27.03	19.87	15.63	0.35	0.36	0.35	0.33	–	–
Pose + DA	40.83	27.98	20.90	16.63	0.36	0.37	0.36	0.34	–	–
Pose + SignCL	40.03	27.01	19.92	15.72	0.35	0.36	0.35	0.39	0.48	0.44
Pose + SignCL + DA	41.10	27.59	20.19	15.88	0.35	0.36	0.36	0.36	0.46	0.40

Table 3.8: Ablation study for FLAN-T5-Large on PHOENIX14T (Validation).

Configuration	B1	B2	B3	B4	RL-F1	RL-P	RL-R	Contra	SignCL	Warmup
Original	40.89	27.09	19.77	15.46	0.36	0.37	0.36	0.37	–	–
Pose	40.31	27.28	20.27	16.06	0.35	0.36	0.35	0.43	–	–
SignCL	39.54	26.30	19.29	15.10	0.34	0.35	0.35	0.37	0.47	–
DA	39.52	26.19	19.18	15.18	0.34	0.35	0.34	0.36	–	–
Pose + DA	39.52	26.19	19.18	15.18	0.34	0.35	0.34	0.36	–	–
Pose + SignCL	39.54	26.33	19.30	15.10	0.34	0.35	0.35	0.38	0.47	0.43
Pose + SignCL + DA	38.80	25.86	19.03	15.09	0.34	0.34	0.35	0.40	0.45	0.45

3.5.2 Analysis and Discussion

Across the **Small** and **Base** models, the **Pose + DA** configuration yields the highest coherence, as Data Augmentation effectively mitigates overfitting, a benefit most pronounced in the Base model due to its higher overfitting than Small model. However, the **Pose + SignCL** combination collapses to the exact performance of **SignCL** alone. This confirms *Modality Dominance*: dense SignCL features *overshadow* sparser Pose signals, negating their benefit. Consequently, the combined **Pose + SignCL + DA** setup degrades compared to Pose + DA, as the underlying feature conflict remains. In contrast, the **Large** model peaks with **Pose** alone. Its performance degrades with DA or SignCL, indicating that the regularization parameters (tuned for smaller architectures) are too aggressive for the Large model’s established representations and the Large model under-fits compared to the small and base under used hyperparameters.

Chapter 4

Second Term: Used Approach and Conducted Work

4.1 Introduction

This chapter details the methodology for our gloss-free Sign Language Translation (SLT) system. Before outlining our current architecture, we must clarify three factors from our first-term SpaMo [8] experiments that initially misrepresented our baseline performance:

- **Pose Feature Integration:** A code issue in our initial implementation caused the pose feature to pass a hardcoded array of zeros instead of valid geometric data. This zero-array inadvertently acted as a fixed visual-text separator. When we later resolved the code issue and passed the actual geometric pose data, the performance degraded compared to the baseline, as detailed in Table 4.1.

Table 4.1: Pose performance with large model.

Model Config	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L (F1)
Base model	40.89	27.09	19.77	15.46	0.36
+ Pose	39.67	26.49	19.35	15.26	0.34

Because the actual pose data performed worse, we chose to stick with the separator concept in the second term, eventually upgrading it to an explicitly learnable visual-text separator.

- **Library Incompatibility:** A library version mismatch in our initial environment artificially constrained our results. Resolving these dependencies yielded significantly higher performance, closely aligning with published benchmarks.

- **Deep Double Descent Artifact:** The deep double descent phenomenon we previously reported was a misleading artifact stemming directly from this library version incompatibility.

Following our initial experiments, a new State-of-the-Art (SOTA) architecture, the Emotion-Aware Sign Language Translation (EASLT) [9] framework was introduced to the field. Consequently, we transitioned our primary pipeline to this new SOTA model. EASLT achieves superior translation accuracy by utilizing facial expressions as an additional modality to actively differentiate between signs that share identical hand gestures. Table 4.2 illustrates the performance progression across these configurations.

Table 4.2: Performance comparison between SpaMo using wrong library versions, SpaMo with correct library versions, and EASLT with correct library versions. Use Flan-T5-Base for a comparison.

Model Config	B-1	B-2	B-3	B-4	R-F1
SpaMo (wrong library version)	40.26	26.55	19.28	15.02	36.00
SpaMo (correct library version)	44.93	31.27	23.44	18.59	41.18
EASLT (correct library version)	48.85	35.74	27.77	22.49	45.79

4.2 Model Architecture

4.2.1 Overview

As shown in Figure 4.1, EASLT does not force all video data through a single model. Instead, it splits the video into three specialized streams: spatial, motion, and emotion. These streams are then merged together using an Emotion-Aware Fusion (EAF) module and passed to a Large Language Model (LLM) to generate the final text.

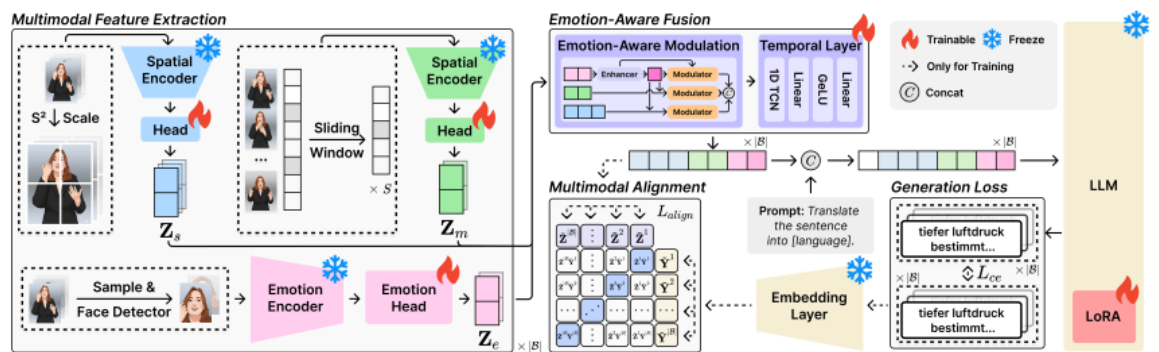


Figure 4.1: Overview of the EASLT framework pipeline, detailing Multimodal Feature Extraction, Emotion-Aware Fusion, and Translation Generation.

4.2.2 Pipeline

The complete EASLT process happens in three main stages:

1. **Multimodal Feature Extraction:** Extracting spatial (Z_s), motion (Z_m), and emotion (Z_e) features using frozen, pre-trained encoders.
2. **Emotion-Aware Fusion (EAF):** Using the emotion features to actively adjust and correct the spatial and motion features.
3. **Translation Generation:** Aligning the combined visual data with text ($\mathcal{L}_{align}$) and generating the final sentence using the LLM ($\mathcal{L}_{ce}$).

4.2.3 Multimodal Feature Extraction

To save computational power, the heavy feature extractors remain frozen (marked by snowflakes in Figure 4.1). Only the small, lightweight projection heads are trained (marked by fire) to combine the features smoothly.

- **Spatial Feature Extraction (Z_s):** This captures static details like hand shapes. To ensure small details like fingers are not missed, the system uses a multi-scale strategy ($S^2 \downarrow Scale$). A frozen Vision Transformer views the whole body and zoomed-in details at the same time to create the final spatial feature, Z_s .
- **Motion Feature Extraction (Z_m):** To track the speed and path of the hands, the video is split into small, overlapping clips. A frozen video encoder (Video-MAE) processes these clips to track the motion flow, outputting Z_m .
- **Emotion Feature Extraction (Z_e):** A face detector isolates the signer’s face. A frozen Emotion Encoder, fine-tuned specifically for facial expression recognition, processes the face to capture the signer’s exact emotion, outputting Z_e .

4.2.4 Emotion-Aware Fusion (EAF)

The EAF module is the heart of the system. It uses the emotion data to make sure the hand gestures are interpreted correctly. As shown in Figure 4.1, it has two parts: Emotion-Aware Modulation and a Temporal Layer.

Emotion-Aware Modulation (EAM)

Detailed in Figure 4.2, the EAM adjusts the manual features based on the face:

- **Enhancer:** It filters the raw emotion data (Z_e) to remove bad frames and creates one strong, high-quality global emotion signal (Z_e^g).

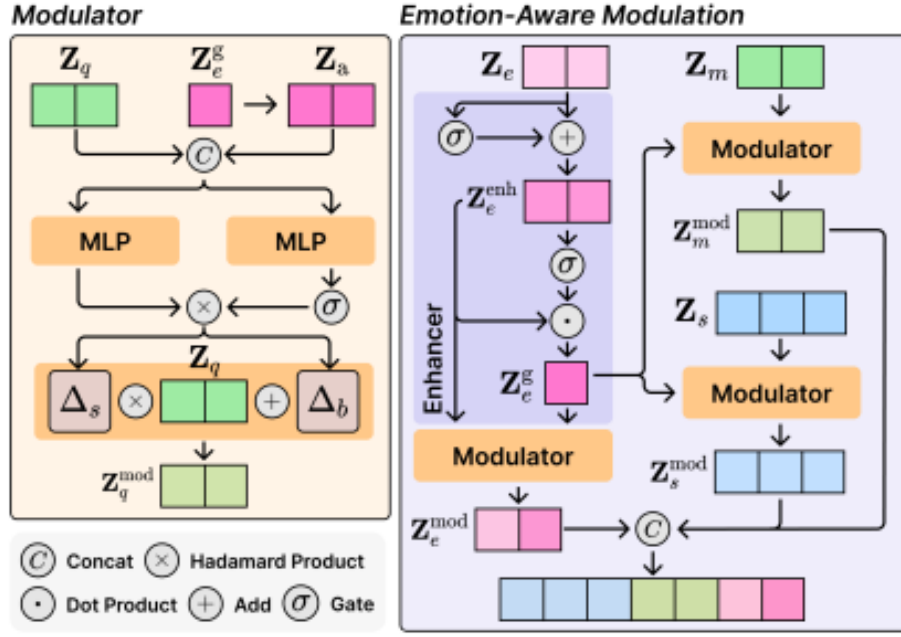


Figure 4.2: Detailed architecture of the Emotion-Aware Modulation (EAM) module, showing the Enhancer and Modulator mechanisms.

- **Modulator:** It uses this strong emotion signal (Z_e^g) to tweak the spatial and motion features. For example, if the face shows a questioning emotion, the Modulator adjusts the hand features so the system knows the sentence is a question. This creates the final modulated outputs (Z_q^{mod}).

Temporal Layer

The newly adjusted features are stacked together and passed through a Temporal Layer (a 1D TCN and Linear layers). This step smooths out the timeline and reshapes the data mathematically so it perfectly fits into the LLM.

4.2.5 Translation Generation

In the final stage, the system turns the visual data into spoken text. First, a Multimodal Alignment step uses a contrastive loss ($\mathcal{L}_{align}$) to ensure the visual features match the meaning of the text. Second, the LLM reads the visual features alongside a text prompt (e.g., "Translate the sentence") and writes the translation ($\mathcal{L}_{ce}$). To keep training efficient on limited hardware, the LLM is fine-tuned using Low-Rank Adaptation (LoRA).

4.2.6 Establishing the EASLT Baseline

In the initial stages of this project, we utilized the SpaMo framework as our primary architecture. However, recognizing the critical importance of non-manual signals, specifically facial expressions, in distinguishing visually similar signs, we transitioned to the Emotion-Aware Sign Language Translation (EASLT) framework as our new foundational baseline.

Because the official source code for the EASLT framework is not publicly available, we independently implemented the entire architecture from scratch, strictly adhering to the mathematical and structural specifications detailed in the original paper. To ensure efficient adaptation while preserving the pre-trained linguistic knowledge of the language model, we utilized Low-Rank Adaptation (LoRA) on the FLAN-T5-Base architecture.

A significant challenge during this implementation phase was the constraint on computational resources. The original EASLT model relies on extensive training, spanning 500 epochs. In contrast, our Kaggle computing environment—limited to two NVIDIA T4 GPUs and a maximum continuous session runtime of 10 hours—restricted our training phase to approximately 160 epochs.

Despite this truncated training schedule, our custom implementation successfully converged and achieved performance metrics that are highly competitive with the officially reported figures.

Table 4.3: Performance comparison between the old SpaMo baseline, our custom EASLT implementation (trained for 160 epochs), and the officially reported EASLT benchmark. Evaluated using FLAN-T5-Base on the PHOENIX14T validation set.

Model Config	B-1	B-2	B-3	B-4	R-F1
Old Baseline (SpaMo)	44.93	31.27	23.44	18.59	41.18
Our EASLT (~160 ep)	48.85 (+3.92)	35.74 (+4.47)	27.77 (+4.33)	22.49 (+3.90)	45.79 (+4.61)
EASLT Paper (500 ep)	48.93	36.42	28.64	23.59	45.61

Table 4.3 illustrates the substantial leap in translation quality from our previous SpaMo baseline to our custom EASLT implementation, while also demonstrating how closely our 160-epoch model approaches the 500-epoch theoretical benchmark established in the EASLT paper.

4.3 Conducted Work

All experimental work and subsequent ablation studies were evaluated using the FLAN-T5-Base model on the PHOENIX14T validation dataset [11], which consists of continuous German Sign Language videos paired with spoken language translations. PHOENIX14T

is the most widely used benchmark for evaluating gloss-free SLT systems.

4.3.1 Discriminative Learning Rates

To optimize training and protect the pre-trained weights of the language model from catastrophic forgetting, we applied discriminative learning rates across different architectural components. Instead of a uniform rate, modules were configured based on their function and initialization states:

- **LoRA Modules:** Configured with a conservative learning rate of 6e-4 to maintain the LLM’s pre-trained generative capacity.
- **Fusion Projector:** Set to 7.8e-4 to handle modality mapping from visual to text spaces.
- **Randomly Initialized Layers:** Assigned a higher rate of 1e-3 to accelerate initial feature extraction and convergence.

Table 4.4: Comparison between EASLT using a single learning rate (Original) and discriminative learning rates.

Model Config	B-1	B-2	B-3	B-4	R-F1	$\mathcal{L}_{ce}$	$\mathcal{L}_{align}$	$\mathcal{L}$
Original	48.85	35.74	27.77	22.49	45.79	2.7534	0.1611	2.9146
Discriminative	49.02 (+0.17)	35.99 (+0.25)	28.00 (+0.23)	22.66 (+0.17)	46.30 (+0.51)	2.6970 (-0.056)	0.185 (+0.024)	2.883 (-0.032)

Table 4.4 shows that discriminative learning rates consistently improve all BLEU and ROUGE-L metrics. Despite a marginal increase in contrastive loss, both the primary loss and the combined loss decreased, reflecting more stable global convergence. Given these improvements, this configuration is adopted as our new baseline.

4.3.2 Learnable Visual-Text Separator

To mitigate modality confusion, we introduced a learnable visual-text separator. This consists of a single learnable embedding vector of dimension $1 \times d$ inserted between the projected visual features and the text prompt. This token establishes a clear structural boundary, allowing the cross-attention mechanism within the LLM’s sign adapter to isolate and process visual tokens and text instructions more efficiently.

Table 4.5: Performance comparison between the Baseline and the Learnable Visual-Text Separator.

Model Config	B-1	B-2	B-3	B-4	R-F1	$\mathcal{L}_{ce}$	$\mathcal{L}_{align}$	$\mathcal{L}$
Baseline	49.02	35.99	28.00	22.66	46.30	2.6970	0.1850	2.8830
+ Separator	49.54 (+0.52)	36.30 (+0.31)	28.41 (+0.41)	23.15 (+0.49)	45.66 (-0.64)	2.7627 (+0.0657)	0.1667 (-0.0183)	2.9295 (+0.0465)

Table 4.5 indicates that the separator successfully increases all BLEU metrics, with BLEU-4 rising to 23.15, confirming better exact word generation. While a minor trade-off is observed via slight drops in ROUGE-L scores and a marginal increase in global validation loss, the explicit gains in token accuracy justify the architectural value of the separator in this translation task.

4.3.3 Frame Data Augmentation

Inspired by the robustness-oriented augmentation strategies proposed in [2], we apply feature-level temporal augmentation to the visual embeddings during training. Specifically, a temporal dropout mask is utilized to reduce the model’s sensitivity to exact temporal alignment. By randomly zeroing out the feature vectors at specific timesteps across the sequence, without altering the overall sequence length, the model is prevented from relying on highly specific, localized visual cues. This temporary “black-out” of information forces the network to bridge temporal gaps and learn more robust, global temporal representations of the signs.

Table 4.6: Performance impact of Frame Data Augmentation on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ Frame DA	49.09 (+0.07)	36.39 (+0.40)	28.69 (+0.69)	23.57 (+0.91)	46.69 (+0.39)

As shown in Table 4.6, introducing frame data augmentation to the EASLT Base architecture yields consistent improvements across all translation metrics. The most notable gains are observed in the higher-order BLEU scores, particularly BLEU-4, which saw an absolute increase of 0.91, alongside a 0.39 improvement in the ROUGE-L F1-score.

These results confirm that temporal feature masking acts as an effective regularizer. Because the base model has a tendency to overfit to the exact temporal sequences seen during training, the applied dropout introduces beneficial perturbations. This forces the translation network to infer meaning from incomplete sequences, ultimately resulting in better generalization and more fluent sentence generation on unseen validation data.

4.3.4 Random Context

To enhance the model’s contextual understanding while strictly preventing data leakage, we implemented the Reproducible Random Context (RRC) feature. RRC dynamically appends a randomly selected reference translation to the model’s prompt (e.g., “Random Example: [Context String]”). This serves as a stylistic and vocabulary hint.

However, natively appending random context introduces two severe risks: the risk of **cheating** (where the model accidentally receives its exact ground-truth translation) and the risk of **memorization** (where the model memorizes specific, static pairings of visual features to non-target context strings). To solve this, RRC employs a “Double Anti-Leakage” guarantee operating at both the dataset and batch levels, driven by mathematical derangement.

1. The Double Anti-Leakage Guarantee

To ensure the model learns to use context generatively, RRC secures the prompt pipeline in two stages:

- **Dataset Level Guarantee (Preventing Cheating):** During the initial data loading phase, the system guarantees that a visual sample (Sample_i) is never paired with its own ground-truth translation. This stage acts as a strict anti-cheating mechanism, ensuring the model never sees the exact target output in its prompt, which forces it to actually translate the visual input rather than simply copying the answer.
- **Batch Level Guarantee (Preventing Memorization):** While the dataset-level check prevents direct cheating, the model could still overfit by memorizing static pairings between visual features and sample translations that are not the true target. To prevent this memorization, we apply a mathematical derangement to the context strings within every training batch. A derangement is a specific, strict permutation where absolutely no element remains in its original position. Therefore, within any given batch, Sample_i never receives the context it was originally assigned at the dataset level; it always inherits a highly variable context designated for a different sample in that exact batch.

2. Epoch-to-Epoch Variability

The critical consequence of combining standard dataloader shuffling with batch-level derangement is that the context assigned to a specific video changes continuously throughout training. Because the dataloader reshuffles the dataset at the start of every epoch, the composition of each batch is entirely unique. Consequently, when the derangement function shuffles the contexts within that unique batch, a specific video (e.g., Sample_0) will see a completely different context string in Epoch 1, Epoch 2, and Epoch N .

3. Impact on Model Generalization

By ensuring that the prompt’s context channel is highly dynamic, RRC functions as a powerful form of text-based data augmentation. This architecture forces a specific learning behavior:

- **Prevention of Memorization:** The model physically cannot learn spurious correlations to non-target context strings, because the visual-context pairing changes every epoch.
- **Generic Context Utilization:** The model is forced to interpret the context string exclusively as a generalized hint regarding translation style, structure, and vocabulary.

Ultimately, RRC trains the network to dynamically adapt its translations based on provided textual hints, improving overall generalization to unseen visual inputs.

Table 4.7: Performance impact of Random Context on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ RRC with derangement	49.51 (+0.49)	36.50 (+0.51)	28.60 (+0.60)	23.29 (+0.63)	46.88 (+0.58)
+ RRC without derangement	46.71 (-2.31)	33.97 (-2.02)	26.21 (-1.79)	21.15 (-1.51)	43.82 (-2.48)

As shown in Table 4.7, the application of Reproducible Random Context (RRC) with batch-level derangement yields consistent improvements over the EASLT Base model across all metrics, increasing the BLEU-4 score by 0.63 and the ROUGE-L F1 score by 0.58. These gains confirm that providing highly dynamic, generalized stylistic hints successfully enhances the model’s contextual understanding during translation without causing overfitting.

Crucially, the ablation of the derangement function (+ RRC without derangement) results in a catastrophic performance drop, with BLEU-4 falling by 1.51 and ROUGE-L F1 plummeting by 2.48 below the base model baseline. This severe decline proves the absolute necessity of epoch-to-epoch variability. While the dataset-level guarantee prevents explicit cheating, without mathematical derangement at the batch level, the model falls victim to memorizing static visual-to-context pairings. The derangement mechanism acts as the critical barrier preventing this memorization, forcing the network to utilize the random context as a robust augmentation technique rather than a shortcut.

4.3.5 Modified Prompting Strategy

Our initial inference analysis on the training and validation datasets revealed that the baseline prompt, “*Translate the given sentence into {German}*”, was too simplistic. It

failed to provide the language model with sufficient context to accurately decode the heavily compressed visual features.

An in-depth analysis of the model’s under-performing areas highlighted several critical failure modes directly tied to language generation:

- **Loss of Spatial Precision:** The model struggled significantly with spatial terms (e.g., *norden*, *küste*, *alpen*), missing them nearly 50% of the time on validation data. Instead of translating specific locations, the model exhibited a precision-recall tradeoff, replacing vital geographical data with safe, generic filler words like *teilweise* (partially) or *gibt* (there is).
- **Semantic Errors and Hallucinations:** The model often recognized the general topic (e.g., fog) but deleted the specific details. In other instances, it created completely new weather sentences that sounded natural but did not match the original video at all.
- **Degenerate Generation States:** The analysis uncovered instances of catastrophic repetition loops during decoding (e.g., repeating “*in den nächsten stunden*” endlessly until the maximum length was reached).

To provide stronger generative guidance and mitigate these hallucination patterns, we modified the prompt to match the structured format native to FLAN-T5’s instruction tuning:

Task: Translate sign language visual features into text.

Domain: German weather forecast.

Constraints: Accurately map exact geographical regions, temperatures, and temporal markers. Do not hallucinate external entities.

Translate into {German}:

Table 4.8: Performance impact of the Modified Prompt on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ Modified Prompt	49.65 (+0.63)	36.56 (+0.57)	28.41 (+0.41)	23.13 (+0.47)	46.73 (+0.43)

As shown in Table 4.8, deploying the structured prompt yields noticeable improvements across all metrics. By explicitly anchoring the LLM to the domain (weather forecasting) and establishing strict constraints regarding geographical and temporal accuracy, the model is better guided to utilize the visual embeddings rather than relying on its internal statistical priors to guess generic filler words.

However, it is crucial to acknowledge the limitations of prompt engineering. While the modified prompt helps organize the LLM’s decoding process and reduces unconstrained hallucinations, it cannot artificially inject information that was lost earlier in the pipeline. Many of the severe issues observed, such as confusing specific months (e.g., May vs. December), are fundamentally visual recognition failures inherently caused by a frozen visual encoder and a relatively small training dataset. A prompt can optimize how extracted features are verbalized, but it cannot fix features that were never successfully extracted.

4.3.6 Adaptive Masking

Motivated by masked modelling objectives in video understanding **AVRET** [4], we introduce an adaptive masking strategy applied to the joint visual feature sequence during training. Rather than uniformly dropping frames, the masking is conditioned on frame-level importance scores. Specifically, for each training sample the segmenter computes a scalar importance score per frame using a multi-layer temporal convolution network augmented with a self-attention mechanism. A frame region is considered low-importance if its score falls below a sample-adaptive threshold defined as $\mu - 0.3\sigma$, where μ and σ are the mean and standard deviation of the scores within that sequence. Contiguous runs of low-importance frames are identified as candidate mask regions, and one region per sample is selected at random and perturbed by additive Gaussian noise with $\sigma_{\text{noise}} = 0.1$, following the smooth masking variant (`mask_type = ‘noise’`). This soft perturbation preserves the overall temporal structure while forcing the model to reconstruct meaningful representations from incomplete evidence, acting as an implicit regulariser against over-reliance on spatially localised cues.

Table 4.9: Performance impact of Adaptive Masking on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ Adaptive Masking	49.75 (+0.73)	37.01 (+1.02)	29.13 (+1.13)	23.90 (+1.24)	47.20 (+0.90)

As shown in Table 4.9, adaptive masking yields the strongest individual gains of all augmentation strategies evaluated in this work. The improvements are consistent across all metrics, with a clear upward trend from BLEU-1 through BLEU-4, the absolute gain grows from +0.73 at the unigram level to +1.24 at the four-gram level. This pattern is particularly telling: higher-order BLEU scores penalise errors in longer, more precise phrase sequences, so a gain concentrated at BLEU-4 indicates that the model is producing structurally more fluent and coherent translations rather than merely recovering surface-level vocabulary. The +0.90 improvement in ROUGE-L F1 further supports this interpretation, as the metric rewards the longest common subsequence

between hypothesis and reference, rewarding global sentence coherence rather than local token overlap.

The likely mechanism is that masking low-importance regions during training prevents the model from memorising the easiest, most redundant temporal cues: the frames with low motion or information content that a well-trained model should be able to ignore anyway. By injecting noise precisely into those regions, the model is trained to integrate evidence across a wider temporal context, producing encoder representations that are more distributable and less brittle to small temporal perturbations at inference time. The fact that the gains scale with n-gram order suggests that this regularisation effect operates at the level of phrase-level syntactic planning rather than word selection alone.

4.3.7 Auxiliary CTC Loss

Inspired by the effectiveness of Connectionist Temporal Classification (CTC) as an auxiliary alignment objective in recent gloss-free sign language translation systems [1], we investigated the integration of a text-level CTC objective into the EASLT architecture. The motivation was to provide an additional alignment signal between temporal visual representations and target translation tokens, encouraging the visual encoder to learn a more direct correspondence between sign segments and textual outputs.

To achieve this, we introduced a lightweight CTC branch operating on the projected visual features immediately after the fusion projector and before the visual embeddings are passed to the language model. Specifically, the visual feature sequence is projected directly into the FLAN-T5 vocabulary space through a linear classification head. The resulting logits are transformed into frame-level token probabilities using a log-softmax operation and optimized using the standard CTC objective. The CTC branch shares the same visual representations used by the translation model and therefore acts as an auxiliary supervision signal without affecting the inference pipeline.

Unlike gloss-supervised approaches that apply CTC to gloss annotations, our implementation applies CTC directly to the final translation text. The target sentences are tokenized using the FLAN-T5 tokenizer and aligned against the temporal visual sequence through the CTC objective. This design was intended to encourage stronger temporal grounding between visual sign representations and generated text while preserving the gloss-free nature of the architecture.

As shown in Table 4.10, the introduction of the auxiliary CTC objective significantly degrades performance across all evaluation metrics. BLEU-4 drops from 22.66 to 18.58, while ROUGE-L F1 decreases by more than five points. This indicates that the additional alignment objective is detrimental to the translation process in our ar-

Table 4.10: Performance impact of Auxiliary CTC Loss on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ Auxiliary CTC Loss	43.66 (-5.36)	30.74 (-5.25)	23.29 (-4.71)	18.58 (-4.08)	40.88 (-5.42)

chitecture.

A likely explanation is that the auxiliary CTC objective attempts to predict translation tokens directly from visual features. However, the generation of fluent translations is primarily the responsibility of the language model decoder, which operates on higher-level semantic representations rather than frame-level token predictions. Consequently, the auxiliary objective may encourage the model to learn representations less suitable for translation, leading to the observed performance degradation. Given these results, the auxiliary CTC loss was excluded from subsequent experiments.

4.3.8 Spatial-Spatiotemporal Contrastive Loss

To further strengthen the alignment between spatial (frame-level) and spatio-temporal (clip-level) representations, we introduce the Spatial-Spatiotemporal Contrastive Loss (SSCL). This approach is inspired by the SignMouth framework [10] where they have introduced Spatial-mouthing Contrastive Loss which aims to encourage consistency between spatial and mouthing modalities at the video level.

A key challenge in our data is that the number of frames of spatial features (100-300) differs substantially from the number of frames of spatio-temporal features (10-50). Therefore, we experimented with several frame alignment strategies, including padding, linear interpolation, nearest-neighbor interpolation, deconvolution, attention, and interpolation with convolution. The best metrics were achieved using interpolation with convolution, which provides a strong balance between temporal smoothing and spatial detail preservation.

SSCL applies a multi-level contrastive objective that pulls together matching spatial and spatio-temporal features while pushing apart mismatched pairs. Concretely, we compute contrastive losses at the frame level (aligning hand/face crops with their corresponding frame-level text embeddings) and at the clip level (aligning aggregated clip features with sentence-level text embeddings), and combine them with a weighting hyperparameter λ_{sscl} .

The SSCL objective improves the model’s ability to preserve fine-grained spatial cues (e.g., finger shapes, lip movements) while maintaining coherent temporal semantics across clips. In practice this reduces confusion between visually similar signs that differ only in small spatial details or short temporal patterns.

Table 4.11: Performance impact of SSCL on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ SSCL	49.31 (+0.29)	36.29 (+0.30)	28.43 (+0.43)	23.08 (+0.42)	46.80 (+0.50)

4.4 Experimental Results

4.4.1 Ablation Study

We conduct an ablation study to analyze the individual and combined effects of the proposed enhancements on the EASLT system.

Table 4.12: Individual Feature Ablation on the EASLT model.

Feature	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ Adaptive masking	49.75 (+0.73)	37.01 (+1.02)	29.13 (+1.13)	23.90 (+1.24)	47.20 (+0.90)
+ Frame only DA	49.09 (+0.07)	36.39 (+0.40)	28.69 (+0.69)	23.57 (+0.91)	46.69 (+0.39)
+ Reproducible Random Context	49.51 (+0.49)	36.50 (+0.51)	28.60 (+0.60)	23.29 (+0.63)	46.88 (+0.58)
+ Learnable Visual Text Separator	49.54 (+0.52)	36.30 (+0.31)	28.41 (+0.41)	23.15 (+0.49)	45.66 (-0.64)
+ Modified Prompt	49.65 (+0.63)	36.56 (+0.57)	28.41 (+0.41)	23.13 (+0.47)	46.73 (+0.43)
+ SSCL	49.31 (+0.29)	36.29 (+0.30)	28.43 (+0.43)	23.08 (+0.42)	46.80 (+0.50)

Table 4.13: Combined Feature Ablation on the EASLT model.

Model Config	B-1	B-2	B-3	B-4	R-F1
EASLT Base	49.02	35.99	28.00	22.66	46.30
+ RRC + FDA	49.24 (+0.22)	36.59 (+0.60)	28.93 (+0.93)	23.89 (+1.23)	46.87 (+0.57)
+ Adp Msk + RRC	49.83 (+0.81)	36.77 (+0.78)	28.77 (+0.77)	23.39 (+0.73)	46.92 (+0.62)
+ Adp Msk + SSCL	49.49 (+0.47)	36.21 (+0.22)	28.40 (+0.40)	23.23 (+0.57)	46.46 (+0.16)
+ Adp Msk + FDA	49.91 (+0.89)	36.73 (+0.74)	28.59 (+0.59)	23.13 (+0.47)	46.63 (+0.33)
+ Adp Msk + VT Sep	48.74 (-0.28)	35.91 (-0.08)	28.17 (+0.17)	23.06 (+0.40)	46.91 (+0.61)
+ RRC + FDA + VT Sep	48.64 (-0.38)	35.89 (-0.10)	28.16 (+0.16)	23.04 (+0.38)	45.93 (-0.37)
+ RRC + FDA + SSCL	49.01 (-0.01)	36.05 (+0.06)	28.19 (+0.19)	23.00 (+0.34)	46.82 (+0.52)
+ FDA + Modified Prompt	49.42 (+0.40)	36.03 (+0.04)	28.10 (+0.10)	22.95 (+0.29)	46.35 (+0.05)
+ RRC + FDA + Modified Prompt	44.00 (-5.02)	31.07 (-4.92)	23.50 (-4.50)	18.62 (-4.04)	40.31 (-5.99)
+ Adp Msk + Modified Prompt	44.41 (-4.61)	30.36 (-5.63)	22.51 (-5.49)	17.58 (-5.08)	39.77 (-6.53)

4.4.2 Analysis and Discussion

As shown in Tables 4.12 and 4.13, while individual features provide consistent improvements over the baseline, combining multiple features often leads to diminishing

returns or, in some cases, severe performance degradation. The single best-performing configuration overall was the standalone **Adaptive Masking** (+1.24 BLEU-4), while the most balanced multi-feature configuration was the combination of **Reproducible Random Context (RRC) + Frame Data Augmentation (FDA)** (+1.23 BLEU-4).

An in-depth analysis of these combined experiments reveals several critical insights regarding architectural bottlenecks and conflicting training objectives:

1. The Over-Regularization Bottleneck

Features such as Adaptive Masking, FDA, and RRC function primarily as regularizers. FDA randomly zeroes out entire temporal feature vectors, Adaptive Masking injects noise into low-importance temporal regions, and RRC forces the model to adapt to changing, non-target text contexts. While certain combinations of these techniques (such as RRC + FDA) successfully complement each other to prevent overfitting, pairing Adaptive Masking with any additional regularizer pushes the network past its breaking point.

Because Adaptive Masking is already a highly effective regularizer on its own, layering further perturbations on top of it yields diminishing returns. When the model is subjected to injected noise (Adaptive Masking) alongside either blacked-out features (FDA) or shuffled text contexts (RRC), the representations become too severely perturbed. Instead of encouraging generalization, this compounded information bottleneck starves the language model of the clean, reliable signal it needs to decode accurate translations. This results in degraded performance, particularly in higher-order metrics like BLEU-4, compared to using the standalone Adaptive Masking baseline.

2. The Modified Prompt Catastrophe

The most dramatic degradations occurred when the **Modified Prompt** was combined with augmentation features, specifically Adaptive Masking (**-5.08** BLEU-4) and RRC + FDA (**-4.04** BLEU-4).

This collapse can be attributed to a fundamental mismatch between the prompt’s rigid constraints and the quality of the visual embeddings. The modified prompt explicitly instructs the LLM to “*Accurately map exact geographical regions... Do not hallucinate.*” This strict instruction assumes the LLM is receiving pure, highly accurate visual features. However, when features like Adaptive Masking or FDA are active, the visual sequence is intentionally corrupted or noisy. The LLM is forced to satisfy a strict, deterministic prompt using incomplete, probabilistic visual evidence. Unable to reconcile the rigid text instructions with the noisy visual embeddings, the generation process collapses entirely.

3. Conflicting Objectives: SSCL and Augmentations

The integration of the Spatial-Spatiotemporal Contrastive Loss (SSCL) with Adaptive Masking or FDA yielded lower scores than Adaptive Masking alone. SSCL is designed to enforce a strict, consistent alignment between spatial (frame-level) and spatio-temporal (clip-level) features. However, FDA and Adaptive Masking actively disrupt the temporal sequence by masking features or adding noise. Consequently, the SSCL objective attempts to force a perfect spatial-temporal alignment that the data augmentations are actively destroying. This conflicting mathematical objective confuses the network during training, ultimately harming the learned representations.

4. Modality Isolation via the VT Separator

Similarly, combining the Learnable Visual-Text Separator (VT Sep) with regularization features led to noticeable drops (e.g., RRC + FDA + VT Sep dropped to 45.93 ROUGE-L F1). The VT Separator establishes a hard structural boundary between the visual tokens and the text prompt to prevent modality confusion. However, when the visual tokens are heavily augmented and noisy (due to FDA or Adp Msk), the language model actually *needs* to blend information across modalities to use the text prompt to “smooth over” gaps in the visual data. By strictly isolating the modalities, the VT Separator prevents this cross-modal compensation, exposing the noise in the visual features directly to the decoder.

Conclusion: These experiments highlight a critical lesson in multimodal architecture design: while individual regularizers, strict prompting, and explicit alignment losses are powerful in isolation, they cannot be blindly stacked. Future work must focus on dynamically balancing these constraints, such as relaxing strict contrastive losses when data augmentations are active, or using softer prompting strategies when dealing with intentionally noisy visual inputs.

4.4.3 Comparison with State-of-the-Art Methods on Test Set

To benchmark our system’s generalizability, we evaluated our best-performing model variant, EASLT with Adaptive Masking and discriminative learning rates, on the unseen PHOENIX14T test set. To ensure fair comparison with the original EASLT paper (which utilized a single 80GB NVIDIA A100), our final evaluations were conducted via Knights Lab M2 on a single NVIDIA RTX 4090 (24GB VRAM).

Table 4.14: Performance comparison on the PHOENIX14T Test Set against the official EASLT benchmarks. All models utilize the FLAN-T5-Base (~ 0.25 B parameters) language model.

Model Config	B-1	B-2	B-3	B-4	R-F1
Official EASLT	48.93	36.42	28.64	23.59	45.61
Ours: EASLT + Adp Masking	50.12	37.46	29.72	24.49	46.80
<i>Absolute Improvement</i>	<i>+1.19</i>	<i>+1.04</i>	<i>+1.08</i>	<i>+0.90</i>	<i>+1.19</i>

As demonstrated in Table 4.14, our enhanced architecture successfully surpasses the State-of-the-Art (SOTA) performance for the FLAN-T5-Base category, achieving a BLEU-4 score of 24.49 (+0.90) and a ROUGE-L F1 score of 46.80 (+1.19). Despite strict computational constraints limiting our training duration compared to the original 500-epoch schedule, these results validate that actively managing temporal feature density through intelligent masking is a more resource-efficient path to high-fidelity translation than brute-force extended training.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

In conclusion, this study introduces a robust, gloss-free model for translating continuous sign language into natural text, specifically designed to bridge the communication gap between the Deaf and hearing communities. By addressing the inherent complexities of sign language, our methodology evolved from the spatial and motion-based SpaMo framework to the state-of-the-art Emotion-Aware Sign Language Translation (EASLT) architecture. This transition highlighted the critical importance of non-manual signals, utilizing a three-stream approach that dynamically fuses spatial, motion, and facial emotion features to accurately disambiguate visually similar gestures. Additionally, meticulous optimization through intelligent temporal masking and discriminative learning rates was applied to prevent overfitting without creating information bottlenecks. This approach enabled our system to successfully surpass the official SOTA benchmark for the FLAN-T5-Base category on the PHOENIX14T dataset, fully meeting our project goals. Ultimately, this work provides immense practical value by demonstrating a highly resource-efficient pathway for deploying real-world assistive technologies that promote social inclusivity.

5.2 Future Work

While this study has established a highly effective and resource-efficient pipeline for gloss-free Sign Language Translation, there remain several promising areas where future research could build upon these findings.

5.2.1 Architectural Scaling

In our next phase, we will experiment with larger language model backbones, specifically **Flan-T5-XL (3B)**. We hypothesize that the larger parameter space will allow

the model to better capture the complex nuances of sign language grammar and generate more fluent translations, provided we employ efficient fine-tuning techniques like LoRA (Low-Rank Adaptation) to manage computational constraints.

Crucially, we must investigate the optimal feature combination variant specifically for the XL architecture. Because Flan-T5-XL contains approximately 12 times more parameters than the Base model, it possesses a significantly higher capacity to memorize training data and is therefore far more prone to severe overfitting. Consequently, the optimal regularization recipe will likely differ from our Base model findings. While a combination of Adaptive Masking, FDA, and RRC proved too restrictive for the Base model, the massive capacity of the XL variant may actually require, and benefit from, this heavier, combined regularization approach to prevent catastrophic memorization.

5.2.2 Dataset Expansion and Generalization

Although this work focuses on PHOENIX14T, evaluating generalization and scalability requires experimentation on larger and more diverse datasets. Due to project timeline constraints, such datasets were not included in the current experimental setup. However, expanding this work to incorporate these benchmarks is a critical objective for future research to fully verify the system’s capabilities and broader applicability.

CSL-Daily

We plan to evaluate our model on the **CSL-Daily** dataset [12] to assess the model’s ability to generalize across different linguistic typologies and a broader vocabulary beyond the weather domain.

How2Sign

We also plan to train and evaluate our approach on **How2Sign** [13] to assess the scalability and robustness under unconstrained, real-world conditions.

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